CE-ViT: A Robust Channel Estimator Based on Vision Transformer for OFDM Systems

Fangyu Liu*, Jing Zhang*, Peiwen Jiang*, Chao-Kai Wen†, and Shi Jin*
*National Mobile Communications Research Laboratory, Southeast University
Nanjing 210096, China, Email: {fangyuliu, jingzhang, peiwenjiang, jinshi}@seu.edu.cn
†Institute of Communications Engineering, National Sun Yat-sen University
Kaohsiung 80424, Taiwan, Email: chaokai.wen@mail.nsysu.edu.tw.

GLOBECOM 2023 - 2023 IEEE Global Communications Conference | 979-8-3503-1090-0/23/$31.00 ©2023 IEEE | DOI: 10.1109/GLOBECOM54140.2023.10436847

***Abstract*—Deep learning (DL) has been widely utilized for channel estimation and has resulted in significant performance improvements. However, most existing research only performs training and testing in relatively static scenarios, leading to a serious deterioration in dynamic scenarios. In this paper, we propose a robust channel estimator for orthogonal frequency-division multiplexing (OFDM) systems in dynamic scenarios called channel estimator Vision Transformer (CE-ViT) based on attention mechanism. We perform a patch embedding operation to process data in both the time and frequency domains, addressing the limitations of the attention mechanism in extracting 2D correlations. Additionally, we introduce tokens that reflect channel characteristics into the network to enhance the robustness. Experimental results show that CE-ViT outperforms the state-of-the-art DL-based methods. Moreover, the addition of tokens significantly improves the performance of CE-ViT in dynamic channel conditions.**

***Index Terms*—Channel estimation, deep learning, attention mechanism, Vision Transformer, OFDM.**

I. Introduction

Orthogonal frequency-division multiplexing (OFDM) is considered a key technology for 5G or post-5G wireless networks due to its ability to efficiently utilize the available spectrum and withstand frequency selective fading. In OFDM systems, the accuracy of channel estimation directly affects the subsequent signal detection and demodulation process. Pilots are usually transmitted together with data to obtain the corresponding channel state information (CSI). Since the positions and values of the pilots are known, the receiver can estimate the channels by using the received signals. Linear minimum mean-squared error (LMMSE) is a typical channel estimation algorithm in OFDM systems [1]. By using second-order channel statistics and noise variance as prior information, LMMSE can decrease estimation errors, but the prior information is difficult to be obtained in dynamic scenarios.

In recent years, deep learning (DL) has been widely used in wireless communication systems, such as channel estimation [2]–[5], signal detection [6], and CSI feedback [7]. For channel estimation, [2] regarded the millimeter-wave channels as images and used a convolutional neural network (CNN) to learn the characteristics of these channels. ChannelNet, a combination of super-resolution network and denoising neural network, was proposed in [3] to enhance channel estimation performance. ReEsNet [4] introduced a deep residual network for channel estimation, achieving high performance and low complexity. In [5], a CNN-based channel estimation network was proposed to combine the performance advantages between spatial-frequency and angle-delay domains. However, the performance of the aforementioned CNN-based networks is dependent on the number of layers and filter sizes, resulting in a large number of parameters and high computational complexity.

Attention mechanism and Transformer structure have shown great potential in the fields of natural language processing and computer vision [8], [9]. Recently, some works have introduced attention modules into channel estimation [10]–[13]. In [10], attention mechanism was used to improve the network performance under different signal-to-noise ratios (SNRs). In [11], the attention mechanism was utilized in the network's encoder to learn the importance of different elements of the input, which were then passed to the decoder for further processing. However, most of the aforementioned works focused on channel estimation in relatively static scenarios, and when the channel environment changes, network performance is severely degraded or unable to function properly.

In this study, we propose a robust attention-based OFDM channel estimation network. The input of the network is the channel matrix after least-squares (LS) estimation [1] and bilinear interpolation. The network first performs patch embedding operation on the interpolated channel, similar to Vision Transformer [9], and then employs the attention mechanism to learn the relationship between subcarriers at different times and frequencies. Additionally, we introduce a token module that combines channel characteristics information (such as SNR, maximum Doppler shift, and delay spread) with the attention mechanism. This enables the network to generate different attention maps (dynamic weights determined by inputs) for different inputs and achieve robustness in various environments. Simulation results show that the proposed network can significantly outperform the state-of-the-art DL-based channel estimators, and achieve remarkable robustness to varying channel environments.

This work was supported in part by the National Natural Science Foundation of China (NSFC) under Grants 62261160576, 62341107, and 61921004, in part by the National Science and Technology Council of Taiwan under grant MOST 111-2221-E-110-020-MY3, and in part by the Sixth Generation Communication and Sensing Research Center funded by the Higher Education SPROUT Project, Ministry of Education of Taiwan.

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II. SYSTEM MODEL AND TRADITIONAL METHODS

A. System Model

We consider a single-input single-output OFDM system with N_{t} OFDM symbols and N_{f} subcarriers. The transmitted signal passes through the fading channel. After removing the cyclic prefix and performing the discrete Fourier transform, the received signal $Y(m,n)$ at the m-th OFDM symbol and the n-th subcarrier can be expressed as

$$Y(m,n) = H(m,n)X(m,n) + W(m,n), \tag{1}$$

where $H(m,n)$, $X(m,n)$, and $W(m,n)$ are the channel matrix, transmitted OFDM symbol, and additive white Gaussian noise (AWGN) with zero mean and variance σ_w^2 at the m-th OFDM symbol and the n-th subcarrier, respectively.

Pilots are usually sparsely placed in the time-frequency grid to obtain accurate channel estimation. In the current 5G new radio (NR) standard [14], a resource block (RB) comprises 12 subcarriers in the frequency domain, while a time slot consists of 14 OFDM symbols in the time domain. A typical pilot pattern defined by the 5G NR standard is adopted. Specifically, two consecutive pilot subcarriers are placed every four subcarriers in the frequency domain, with the 3-rd and 12-th OFDM symbols within a time slot serving as pilot symbols in the time domain.

The received signal $\mathbf{Y}_{\mathrm{P}} \in \mathbb{C}^{N_{\mathrm{f,P}} \times N_{\mathrm{t,P}}}$ located at the pilot positions in the frequency domain is expressed as

$$\mathbf{Y}_{\mathrm{P}} = \mathbf{H}_{\mathrm{P}} \odot \mathbf{X}_{\mathrm{P}} + \mathbf{W}_{\mathrm{P}}, \tag{2}$$

where $\mathbf{H}_{\mathrm{P}}$, $\mathbf{X}_{\mathrm{P}}$, $\mathbf{W}_{\mathrm{P}} \in \mathbb{C}^{N_{\mathrm{f,P}} \times N_{\mathrm{t,P}}}$ are the channel matrix, transmitted OFDM symbols, and additive white Gaussian noise of the pilot positions, respectively. $\odot$ is the Hadamard product (element-wise product). $N_{\mathrm{f,\,P}}$ is the number of subcarriers as pilots on the frequency axis, and $N_{\mathrm{t,\,P}}$ is the number of OFDM symbols as pilots on the time axis.

B. Traditional Methods

For LS channel estimation [1], the estimated channel at pilot positions $\widehat{\mathbf{h}}_{\mathrm{P}}^{\mathrm{LS}} \in \mathbb{C}^{(N_{\mathrm{f,P}} N_{\mathrm{t,P}}) \times 1}$ can be obtained by

$$\widehat{\mathbf{h}}_{\mathrm{P}}^{\mathrm{LS}} = \arg\min_{\mathbf{h}_{\mathrm{P}}} \|\mathbf{y}_{\mathrm{P}} - \mathbf{h}_{\mathrm{P}} \odot \mathbf{x}_{\mathrm{P}}\|_2^2 = \mathbf{y}_{\mathrm{P}} \odot \mathbf{x}_{\mathrm{P}}^{\odot -1}, \tag{3}$$

where $\mathbf{x}_{\mathrm{P}}$, $\mathbf{y}_{\mathrm{P}}$, $\mathbf{h}_{\mathrm{P}}$, $\widehat{\mathbf{h}}_{\mathrm{P}}^{\mathrm{LS}} \in \mathbb{C}^{(N_{\mathrm{f,P}} N_{\mathrm{t,P}}) \times 1}$ are vectorized $\mathbf{X}_{\mathrm{P}}$, $\mathbf{Y}_{\mathrm{P}}$, $\mathbf{H}_{\mathrm{P}}$, $\widehat{\mathbf{H}}_{\mathrm{P}}^{\mathrm{LS}}$, respectively. $\widehat{\mathbf{H}}_{\mathrm{P}}^{\mathrm{LS}} \in \mathbb{C}^{N_{\mathrm{f,P}} \times N_{\mathrm{t,P}}}$ is the LS estimation of $\mathbf{H}_{\mathrm{P}}$. $\|\cdot\|_2$ is the ℓ_2 distance. $(\cdot)^{\odot -1}$ is the Hadamard inverse (element-wise inverse). The channel at data positions is obtained by interpolation methods, such as linear interpolation and Gaussian interpolation.

When the noise power and second-order statistical characteristics of the channel are known, the LMMSE channel estimation [1] can be used to improve the estimation accuracy. The LMMSE channel estimation of complete channel $\widehat{\mathbf{h}}_{\mathrm{LMMSE}} \in \mathbb{C}^{(N_{\mathrm{f}} N_{\mathrm{t}}) \times 1}$ can be expressed as

$$\widehat{\mathbf{h}}_{\mathrm{LMMSE}} = \mathbf{R}_{\mathbf{h}\mathbf{h}_{\mathrm{P}}} (\mathbf{R}_{\mathbf{h}_{\mathrm{P}}\mathbf{h}_{\mathrm{P}}} + \sigma_w^2 \mathbf{I})^{-1} \widehat{\mathbf{h}}_{\mathrm{P}}^{\mathrm{LS}}, \tag{4}$$

where $\mathbf{R}_{\mathbf{h}\mathbf{h}_{\mathrm{P}}} = \mathbb{E}\{\mathbf{h}\mathbf{h}_{\mathrm{P}}^{\mathrm{H}}\}$ is the cross-correlation matrix between $\mathbf{h}$ and $\mathbf{h}_{\mathrm{P}}$, $\mathbf{R}_{\mathbf{h}_{\mathrm{P}}\mathbf{h}_{\mathrm{P}}} = \mathbb{E}\{\mathbf{h}_{\mathrm{P}}\mathbf{h}_{\mathrm{P}}^{\mathrm{H}}\}$ is the auto-correlation matrix of $\mathbf{h}_{\mathrm{P}}$. $\mathbf{h} \in \mathbb{C}^{(N_{\mathrm{f}} N_{\mathrm{t}}) \times 1}$ is the vectorized complete true channel $\mathbf{H} \in \mathbb{C}^{N_{\mathrm{f}} \times N_{\mathrm{t}}}$.[1]

III. ATTENTION-BASE CHANNEL ESTIMATION

In this section, we first present the network's structures and the reasons for choosing these structures. Then, we highlight some parameter settings in the network training process.

A. Network Structure

We propose a robust Vision Transformer channel estimation network (CE-ViT) based on the attention mechanism for OFDM systems in varying scenarios, as shown in Fig. 1. CE-ViT mainly consists of four parts: an upsampling module, a patch embedding module, a token module, and a Transformer encoder module. The network uses the attention mechanism to identify important elements during estimation. Additionally, channel characteristics are introduced to enhance the robustness of the network.

1) Upsampling module: Different from the methods in [4], [11], [12] that first perform LS estimation on the pilot positions and then upsample the processed results, we relocate the upsampling module to the front of the transformer encoder module. This operation enables the attention mechanism to grasp the positional relationship of channel elements and enhances channel estimation accuracy.

The upsampling module first calculates the LS estimation $\widehat{\mathbf{H}}_{\mathrm{P}}^{\mathrm{LS}}$ at the pilot positions using (3), and then performs bilinear interpolation on the $\widehat{\mathbf{H}}_{\mathrm{P}}^{\mathrm{LS}}$ to obtain the roughly estimated channel response $\widehat{\mathbf{H}}_{\mathrm{int}} \in \mathbb{C}^{N_{\mathrm{f}} \times N_{\mathrm{t}}}$, which can be expressed as

$$\widehat{\mathbf{H}}_{\mathrm{int}} = f_{\mathrm{int}}(\widehat{\mathbf{H}}_{\mathrm{P}}^{\mathrm{LS}}), \tag{5}$$

where f_{int} represents the bilinear interpolation. Due to the correlation between the real and imaginary parts of $\widehat{\mathbf{H}}_{\mathrm{int}}$, the two parts are concatenated along the time dimension to form a new size denoted as $\mathbb{R}^{N_{\mathrm{f}} \times 2N_{\mathrm{t}}}$ to better capture the variations in the channel.[2]

2) Patch embedding module: Affected by multi-path and Doppler effects, the channel exhibits frequency-selective fading and fast time-varying in the time-frequency grids, as illustrated in Fig. 2. Channel estimation becomes challenging when the channel experiences severe fading. For instance, the channel rapidly changes over time in the presence of a severe Doppler effect. Traditional methods struggle to accurately track the time variations of the channel with limited pilot frequencies in these cases. However, the attention mechanism can effectively capture the correlation between pilots and emulate the time or frequency changes of the channel, leading to an enhanced channel estimation performance. In high-velocity scenarios, as depicted in Fig. 2(a), the authors of

[1]The calculations of $\mathbf{R}_{\mathbf{h}\mathbf{h}_{\mathrm{P}}}$ and $\mathbf{R}_{\mathbf{h}_{\mathrm{P}}\mathbf{h}_{\mathrm{P}}}$ require real channel responses, which are not available in practice. Therefore, we calculate and save $\mathbf{R}_{\mathbf{h}\mathbf{h}_{\mathrm{P}}}$ and $\mathbf{R}_{\mathbf{h}_{\mathrm{P}}\mathbf{h}_{\mathrm{P}}}$ by executing a certain number of channel simulations, and then use them as prior channel information for LMMSE.

[2]The $\widehat{\mathbf{H}}_{\mathrm{int}}$ will be represented in the real-valued form in the subsequent parts for the sake of conciseness.

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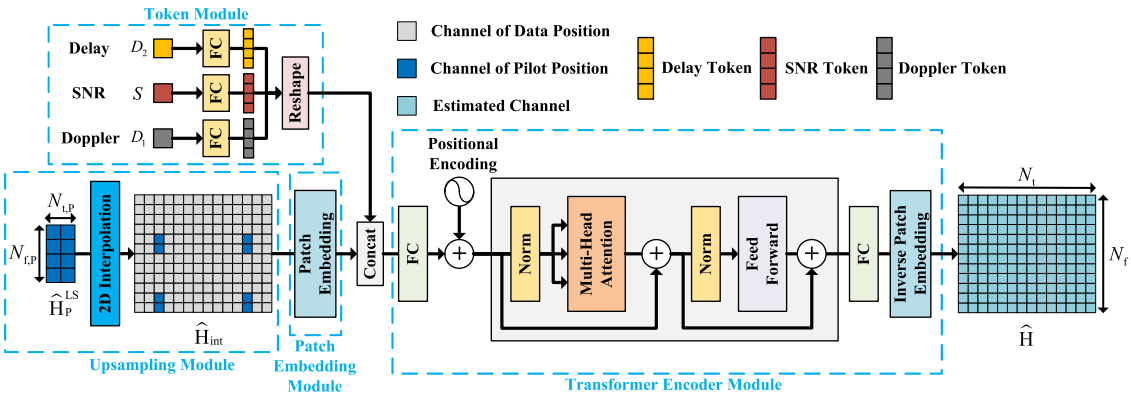
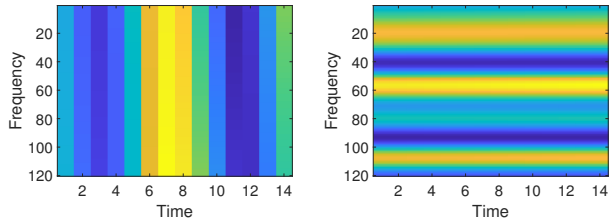


Fig. 1. Architecture of CE-ViT.



(a) Doppler-affected channel. (b) Multi-path affected channel.

Fig. 2. The channel affected by Doppler effect and multi-path effect. (These figures depict the absolute value of the channel response.)

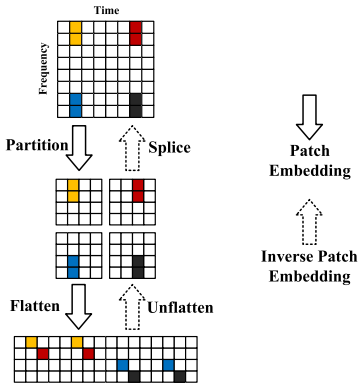


Fig. 3. The process of patch embedding and inverse patch embedding.

[13] utilized Transformer to track channel variation in the time domain and achieved high performance.

However, the attention mechanism can only calculate the correlation along one dimension of the input, resulting in its inability to simultaneously track the time and frequency variations of the doubly selective channels. Therefore, a patch embedding operation similar to [9] is applied to the real-valued matrix $\widehat{\mathbf{H}}_{\text{int}}$ before the Transformer encoder. $\widehat{\mathbf{H}}_{\text{int}}$ is partitioned into patches of fixed size and each patch is flattened into 1D vectors, as illustrated in Fig. 3. The patch size is set to $\mathbb{R}^{10\times 4}$, resulting in a channel response size of $\mathbb{R}^{\frac{N_{\text{f}}N_{\text{t}}}{20}\times 40}$ after the patch embedding module. Through the utilization of the patch embedding operation, each patch incorporates both time-domain and frequency-domain data. When calculating the correlation between patches, the attention mechanism can simultaneously process the data in time and frequency domains, thus learning the correlation in two domains. This allows the attention mechanism to effectively track the overall variations of the channel.

3) Token module: The interpolated channel is characterized by significant parameters such as SNR, maximum Doppler shift, and delay spread. SNR quantifies the level of noise in the interpolated channel, whereas maximum Doppler shift and delay spread reveal the rate of channel variation in the time and frequency domains, respectively. These channel parameters can also be obtained using methods described in [15]–[17]. Once these parameters are obtained, they are fed into the network along with the channel to assist channel estimation.

Traditional neural networks have fixed weights that cannot adapt to input variations. However, the weights of the attention mechanism can be dynamically adjusted based on the input. Therefore, these parameters are integrated with the attention mechanism, allowing the attention mechanism to adapt the attention maps to the input channels of different characteristics, thereby enhancing network performance and robustness. If the network is directly fed with channels of different characteristics without any preprocessing, its performance may be adversely affected.

Three fully-connected structures are separately used to convert the values of SNR, maximum Doppler shift, and delay spread to tokens of size $\mathbb{R}^{\frac{N_{\text{f}}N_{\text{t}}}{10}}$. These tokens are then reshaped to a size of $\mathbb{R}^{\frac{N_{\text{f}}N_{\text{t}}}{20}\times 2}$, and combined with the patched channel response in the second dimension to get the input of the Transformer encoder with size $\mathbb{R}^{\frac{N_{\text{f}}N_{\text{t}}}{20}\times 46}$.

4) Transformer encoder module: The encoder of Transformer is the primary structure of the channel estimator because its attention mechanism can more effectively suppress noise. In contrast, traditional neural networks may extract redundant features, resulting in performance degradation or requiring higher computational complexity to achieve the same performance as the attention mechanism.

The Transformer encoder mainly includes two blocks: multi-head attention and feed-forward network. Firstly, a fully-

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connected layer is used to transform the data from size $\mathbb{R}^{\frac{N_\mathrm{f}N_\mathrm{t}}{20}\times 46}$ to size $\mathbb{R}^{\frac{N_\mathrm{f}N_\mathrm{t}}{20}\times d_\mathrm{model}}$, where d_model is the output dimension of the fully-connected layer. A learnable parameter layer of size $\mathbb{R}^{\frac{N_\mathrm{f}N_\mathrm{t}}{20}\times d_\mathrm{model}}$ is incorporated as the positional encoding to assist Transformer to grasp the position relationship between different patches. Layer normalization [18] is applied before every block, and residual connects after every block. In multi-head attention, the size of input data is first transformed into $\mathbb{R}^{\frac{N_\mathrm{f}N_\mathrm{t}}{20}\times 3d_\mathrm{model}}$ through an internal fully-connected layer. Then, the transformed input is divided into three sub-inputs named query $\mathbf{Q}$, key $\mathbf{K}$, and value $\mathbf{V}$. All of the sub-inputs have the same size as $\mathbb{R}^{\frac{N_\mathrm{f}N_\mathrm{t}}{20}\times\frac{d_\mathrm{model}}{N_\mathrm{head}}\times N_\mathrm{head}}$. Where N_head is the number of heads in multi-head attention. Then these sub-inputs are divided into different heads and each head performs scaled dot-product attention operation as

$$\mathrm{Attention}(\mathbf{Q}_i,\mathbf{K}_i,\mathbf{V}_i)=\mathbf{A}_i\mathbf{V}_i=\mathrm{softmax}\left(\frac{\mathbf{Q}_i{\mathbf{K}_i}^T}{\sqrt{d_\mathrm{k}}}\right)\mathbf{V}_i, \tag{6}$$

where $\mathbf{Q}_i$, $\mathbf{K}_i$, $\mathbf{V}_i\in\mathbb{R}^{\frac{N_\mathrm{f}N_\mathrm{t}}{20}\times\frac{d_\mathrm{model}}{N_\mathrm{head}}}$ are the query matrix, key matrix, and value matrix of i-th head, respectively. $\mathbf{A}_i\in\mathbb{R}^{\frac{N_\mathrm{f}N_\mathrm{t}}{20}\times\frac{N_\mathrm{f}N_\mathrm{t}}{20}}$ is the attention map of i-th head, and higher values in the attention map imply greater contribution from the corresponding positions in the value matrix $\mathbf{V}_i$ to the output. $d_\mathrm{k}=\frac{d_\mathrm{model}}{N_\mathrm{head}}$ is the last dimension of $\mathbf{K}_i$.

The scaled dot-product results of each head are combined and then transferred to the subsequent block. The feed-forward network is a fully-connected network with three layers, where the hidden layer has $2d_\mathrm{model}$ neurons and the output layer has d_model neurons. The GELU activation function [19] is deployed as the activation function between the two layers. At the end of the network, a single-layer fully-connected network is used to remap the output dimension to $\mathbb{R}^{\frac{N_\mathrm{f}N_\mathrm{t}}{20}\times 40}$. The final prediction result $\widehat{\mathbf{H}}\in\mathbb{C}^{N_\mathrm{f}\times N_\mathrm{t}}$ is obtained by performing the inverse patch embedding operation (as shown in Fig. 3) and splicing together the real and imaginary parts.

B. Model Training

The set of network parameters is denoted as $\mathbf{\Theta}$. The input to CE-ViT consists of the LS estimation at pilot positions $\widehat{\mathbf{H}}_\mathrm{P}^\mathrm{LS}$, SNR information S, maximum Doppler shift information D_1, and delay spread information D_2. The final estimated channel response $\widehat{\mathbf{H}}$ can be defined as

$$\widehat{\mathbf{H}}=f(\mathbf{\Theta};f_\mathrm{patch}(f_\mathrm{int}(\widehat{\mathbf{H}}_\mathrm{P}^\mathrm{LS})),S,D_1,D_2), \tag{7}$$

where f is the CE-ViT function, f_patch is the patch embedding functions.

We use the mean-squared error (MSE) loss function to evaluate the error between the estimated channel response $\widehat{\mathbf{H}}$ and the true channel response $\mathbf{H}$, yielding

$$L_1=\frac{1}{N}\sum_{j=1}^{N}\left\|\mathbf{H}_j-\widehat{\mathbf{H}}_j\right\|_2^2, \tag{8}$$

where $\mathbf{H}_j$ is the j-th true channel response in training dataset, N is the total number of training samples.

IV. SIMULATION RESULTS

In this section, we perform simulations on the proposed network to validate its performance. An OFDM system with $N_\mathrm{f}=120$ subcarriers and $N_\mathrm{t}=14$ symbols is considered. Quadrature phase-shift keying is used as the modulation method. The number of pilots in frequency and time domains are $N_\mathrm{f,P}=40$ and $N_\mathrm{t,P}=2$, respectively. The simulated channels are based on the TDL-A model from the 3GPP tapped delay line model [20]. The carrier frequency is set to 3.5 GHz with a subcarrier spacing of 15 kHz. The delay spread is selected from $[25,50,...,300]$ ns, the maximum Doppler shift is selected from $[50,100,...,1000]$ Hz, and the SNR is selected from $[0,5,...,25]$ dB. The training dataset contains 102,400 independent channels, with 90% used for training and 10% used for validation. The channel parameters of the test dataset are selected according to the corresponding scenarios, and each test point contains 2000 test samples. All the networks are trained and tested in Python 3.7 and PyTorch 1.8.1.

The adaptive moment estimation (Adam) optimizer is used to train our network, with an initial learning rate, number of epochs, and batch size of 0.001, 1000, and 512, respectively. The learning rate exponentially decays every 500 epochs by a factor of 10.

We evaluate the performance of several models, including CE-ViT, a variant of CE-ViT with token removal (ViT), and a baseline ViT network (Baseline) trained and tested under corresponding SNR, maximum Doppler shift, and delay spread channels. To strike a balance between performance and complexity, d_model and N_head are set to 128 and 4, respectively. Additionally, we compare the proposed network with traditional and other DL-based methods, such as LS, LMMSE, ChannelNet [3], ReEsNet [4], and HA02 [11]. The hyperparameters of ReEsNet and HA02 are fine-tuned to adapt to the differences in the datasets .[3]

A. Performance Analysis of MSE Under Various Channel Conditions

In this section, we investigate the relationship between the network performance and SNR, maximum Doppler shift, and delay spread, aiming to comprehensively evaluate the performance of each network. Fig. 4 shows the MSE performance versus SNR for each method. The test dataset is generated in the TDL-A channel with maximum Doppler shift of 500 Hz and delay spread of 200 ns. In this figure, LS performs the worst as it ignores the effect of the noise and the channel of data position can only be obtained through interpolation. LMMSE outperforms LS at low SNRs by utilizing noise variance and second-order statistics, but its performance approaches LS as SNRs increase. The performance of ReEsNet and HA02 exceeds that of ChannelNet and LMMSE. CE-ViT outperforms other methods and maintains good performance

[3]ReEsNet is configured with 6 ResBlocks. The last convolutional layer has 2 filters of size 5×5, while the remaining convolutional layers have 16 filters of size 7×7. HA02 has 3 residual convolutional blocks, the last convolutional layer has 1 filter of size 5×5, and the other convolution layers have 16 filters of size 5×5.

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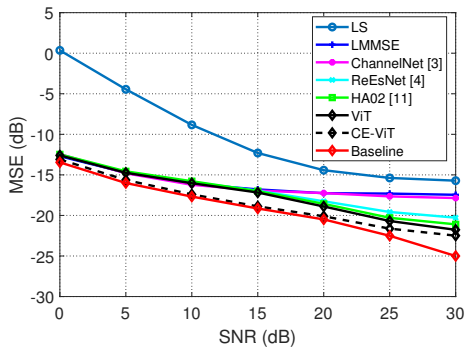

Fig. 4. MSE performance versus SNR.

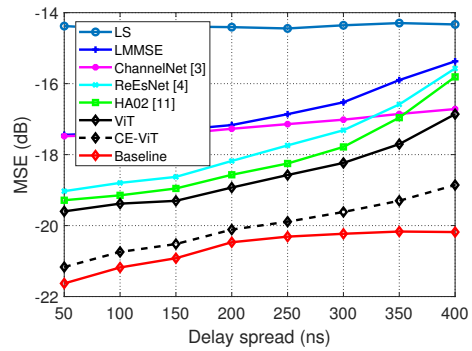

Fig. 6. MSE performance versus delay spread.

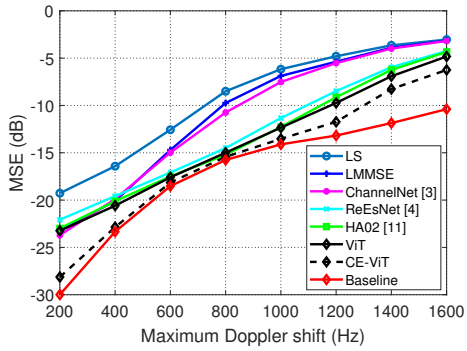

Fig. 5. MSE performance versus maximum Doppler shift.

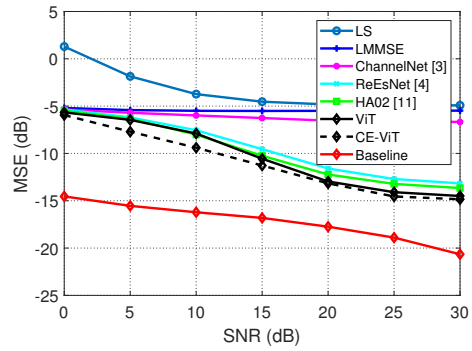

Fig. 7. Robustness analysis of each algorithm.

even in the absence of tokens (ViT). With the addition of tokens, its performance significantly improves, approaching that of the network trained and tested on the same datasets. Furthermore, CE-ViT performs well in the case of SNR mismatch, such as 30 dB.

Fig. 5 illustrates the relationship between the performance of each method and the maximum Doppler shift. The test dataset is generated in the TDL-A channel with delay spread of 200 ns and SNR of 20 dB. The results show that LS still performs the worst. LMMSE has an advantage at low maximum Doppler shifts but deteriorates as the maximum Doppler shift increases. The performance of ChannelNet remains close to that of LMMSE, while ReEsNet exhibits a performance level comparable to that of HA02. The proposed method outperforms other DL-based methods in most cases, even without adding tokens (ViT). The network performance shows a significant improvement with the addition of tokens, yielding a performance gain of approximately 5 dB under a maximum Doppler shift of 200 Hz. Furthermore, the network operates effectively in the environments with maximum Doppler shift mismatches ranging from 1000 Hz to 1600 Hz.

Fig. 6 shows the performance of each method under varying delay spread. The test dataset is generated in the TDL-A channel with maximum Doppler shift of 500 Hz and SNR of 20 dB. The results demonstrate that the performance of each method has a certain downward trend with the increase of delay spread. However, the proposed network is superior to other DL-based methods, and CE-ViT presents obvious performance improvement compared to ViT without introducing tokens. Furthermore, CE-ViT does not exhibit significant performance degradation in the presence of delay spread mismatches.

B. Analysis of Robustness and Performance Gain

DL-based channel estimators are typically trained under fixed datasets that may not match the application scenarios (e.g., channel model, delay spread) and lead to performance degradation. Fig. 7 shows the robustness of each method under environmental mismatches of the TDL-E channel model, 400 ns delay spread, and 1200 Hz maximum Doppler shift. LS and LMMSE perform poorly due to severe Doppler effect, while DL-based methods are also affected by data mismatch. CE-ViT demonstrates the least performance degradation, indicating its great robustness compared to other methods.

Furthermore, Fig. 8 demonstrates the effectiveness of adding tokens for performance improvement. This figure illustrates the attention maps of four heads in both ViT and CE-ViT under two distinct channel conditions. Here, Channel 0 represents the interpolated channel with SNR of 25 dB, maximum Doppler shift of 100 Hz, and delay spread of 300 ns. Channel 1 represents the interpolated channel with SNR of 25 dB, maximum Doppler shift of 1000 Hz, and delay spread of 50 ns. Heads 0 to 3 represent different heads in the multi-head attention.

The blackness of each point increases with its value in the attention maps. As illustrated in the figure, the attention maps of ViT exhibit minimal variations during the transition

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from Channel 0 to Channel 1, while the attention maps of each head in CE-ViT undergo significant changes. Introducing tokens enables the network to adapt to different channels and improves generalization.

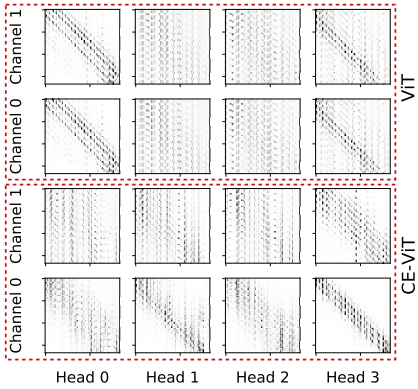

Fig. 8. Attention maps of ViT and CE-ViT.

C. Complexity Analysis

We compare the complexity of CE-ViT and ViT with other channel estimators, including LS, LMMSE, ChannelNet, ReEsNet, and HA02. Table I shows that the calculation amount for LS is low because it only involves complex division of pilot positions and linear interpolation of data positions. However, LMMSE involves matrix multiplication and inversion, resulting in high computational complexity. ChannelNet and ReEsNet have multiple layers of convolutional structures, resulting in higher computational costs compared to other methods. ViT and CE-ViT offer obvious advantages in terms of training parameters and computational complexity, and their performance outperforms the other networks. Furthermore, CE-ViT only incurs a slight increase in the number of parameters and computational complexity, yet yields a significant performance gain compared to ViT.

TABLE I
COMPLEXITY ANALYSIS

Algorithm	Parameters	FLOPs
LS [1]	-	0.044M
LMMSE [1]	-	136.78M
ChannelNet [3]	0.68M	2299.89M
ReEsNet [4]	0.18M	71.18M
HA02 [11]	0.22M	23.92M
ViT	0.14M	24.13M
CE-ViT	0.23M	24.42M

V. CONCLUSION

In this paper, we propose an attention-based channel estimation network called CE-ViT. The network utilizes the attention mechanism to extract information from inputs and improve the accuracy of channel estimation. Channel characteristics, such as SNR, maximum Doppler shift, and delay spread, are introduced into the network in the form of tokens, and learning is carried out by the attention mechanism, both of which further improve the robustness of the network. In comparison to other DL-based methods, CE-ViT has low computational complexity. In addition, simulation results demonstrate that the proposed network exhibits significant performance improvements and remarkable robustness to varying channel environments.

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